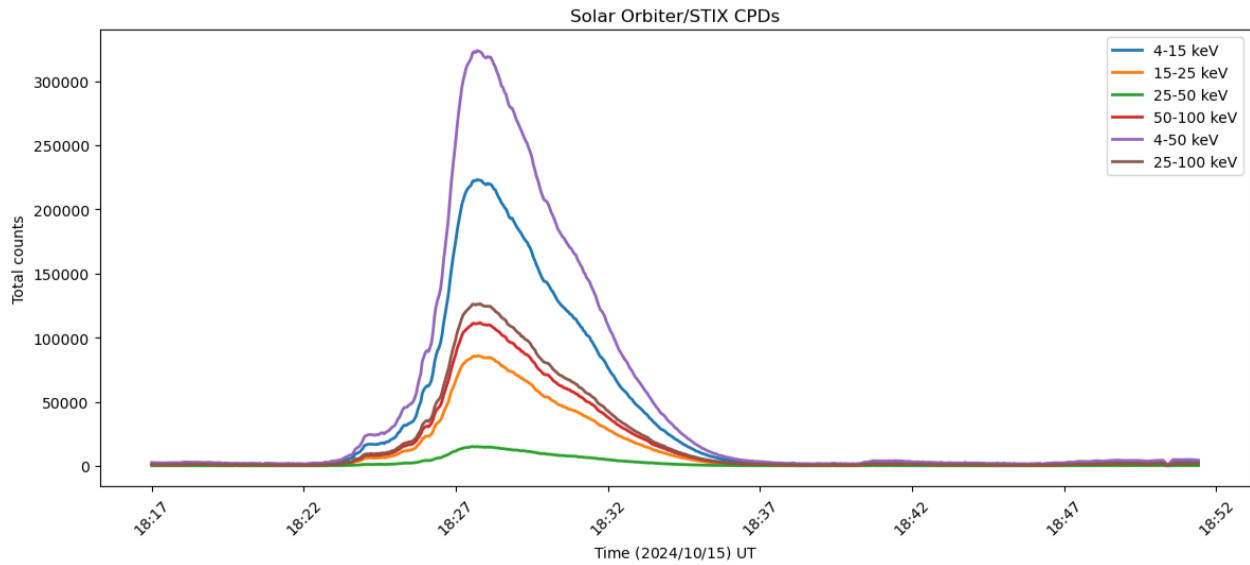


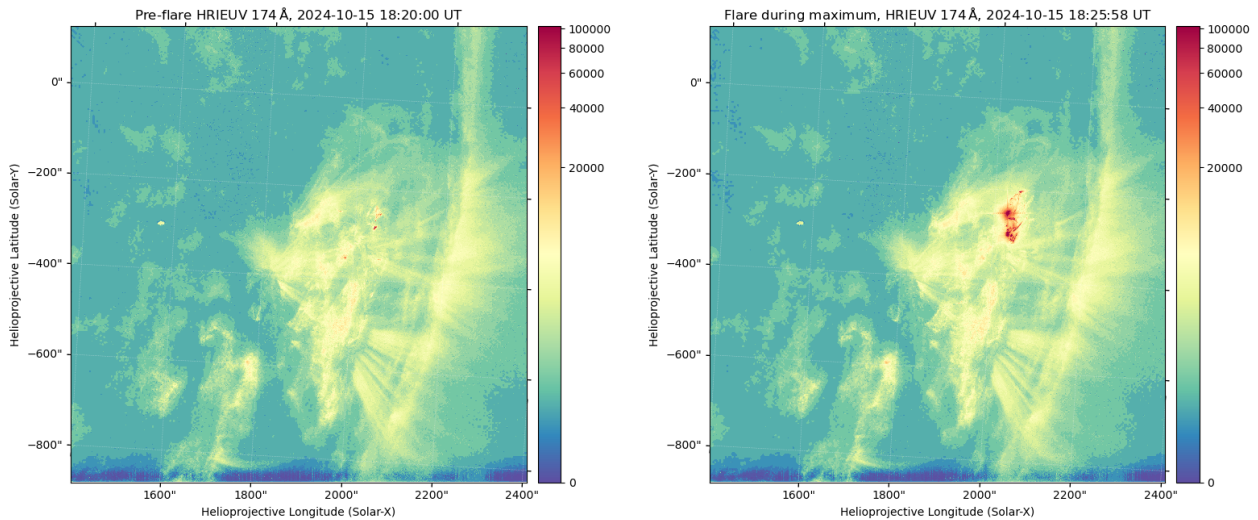
Summer student internship: Multiwavelength Analysis of Quasi-Periodic Pulsations (QPPs) in a Solar Flare | **PROGRESS REPORT no. 2** | 01/08/2025

For the second progress report we analyzed M2.0-class solar flare (occured on 15/10/2024 around 18 : 30 UT), which was observed during the Solar Orbiter Major Flare Watch Campaign. Analysis was done through multiple steps, and here we're showing key plots and conclusions.

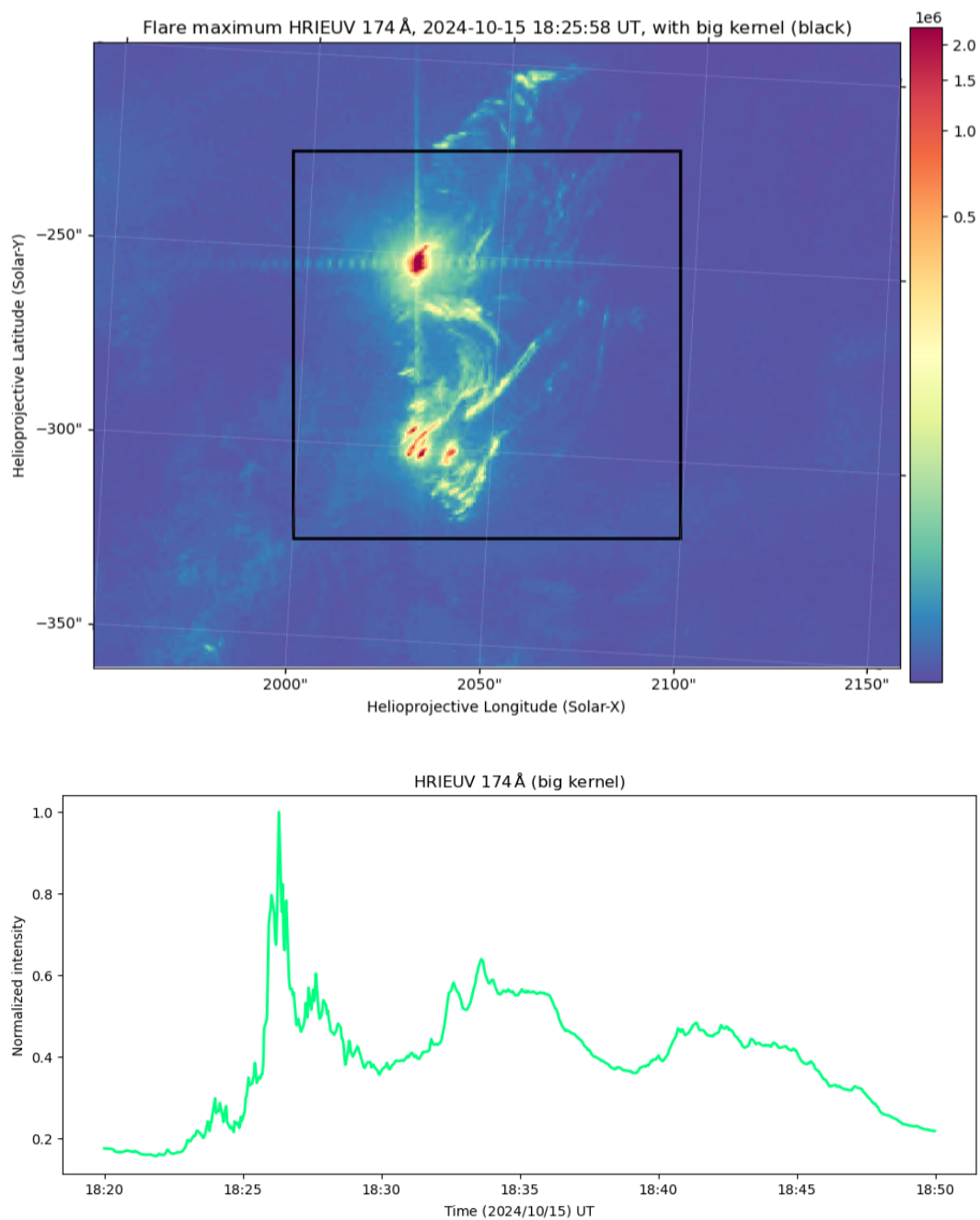
Bellow, we show Compressed Pixel Data (CPD) from STIX - this format is more suitable for general examination and detailed QPP analysis.



Furthermore, we show the observed region ($\lambda = 174 \text{ \AA}$) for pre-flare and flare maximum timestamps.

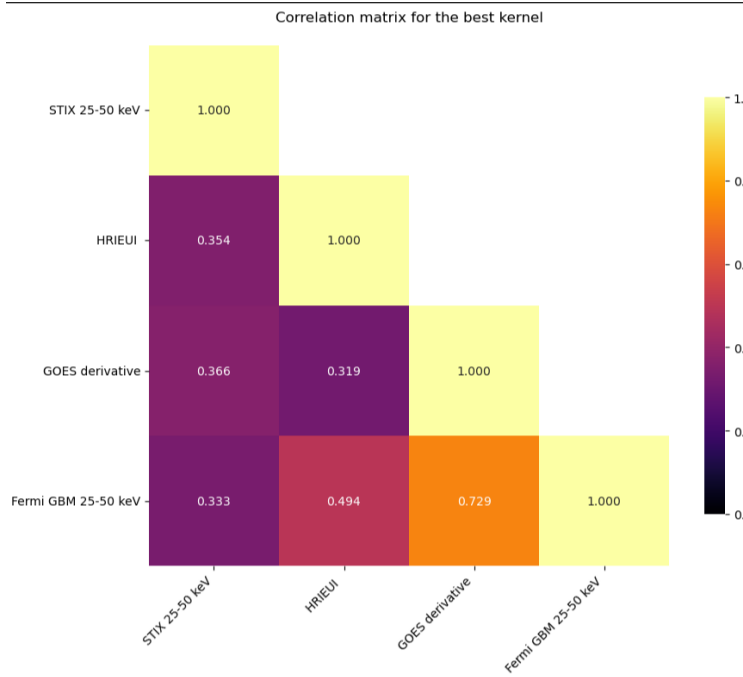
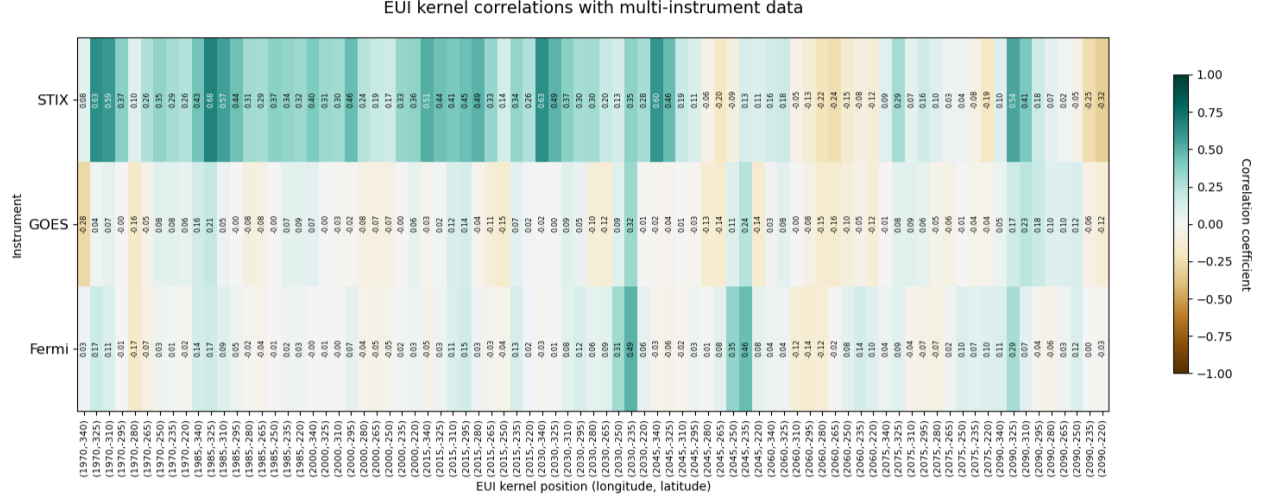


We located one big kernel (black-colored) and extracted light curve from it. This light curve is not time-corrected (we do that later¹, for certain light curves).

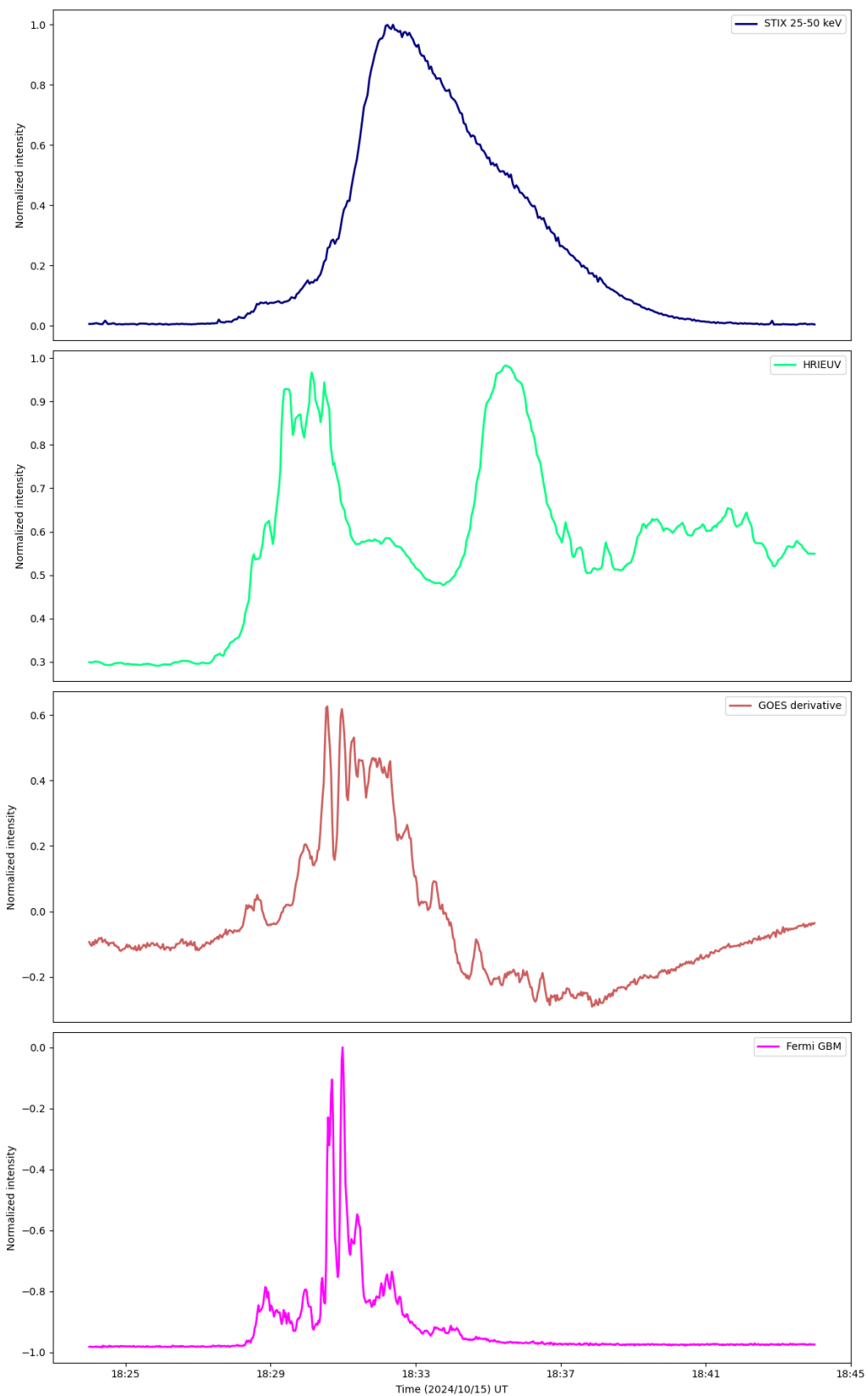


¹There is a time shift between Earth (i.e. GOES/ Fermi) and Solar Orbiter (i.e. STIX/EUI) of around 218 s.

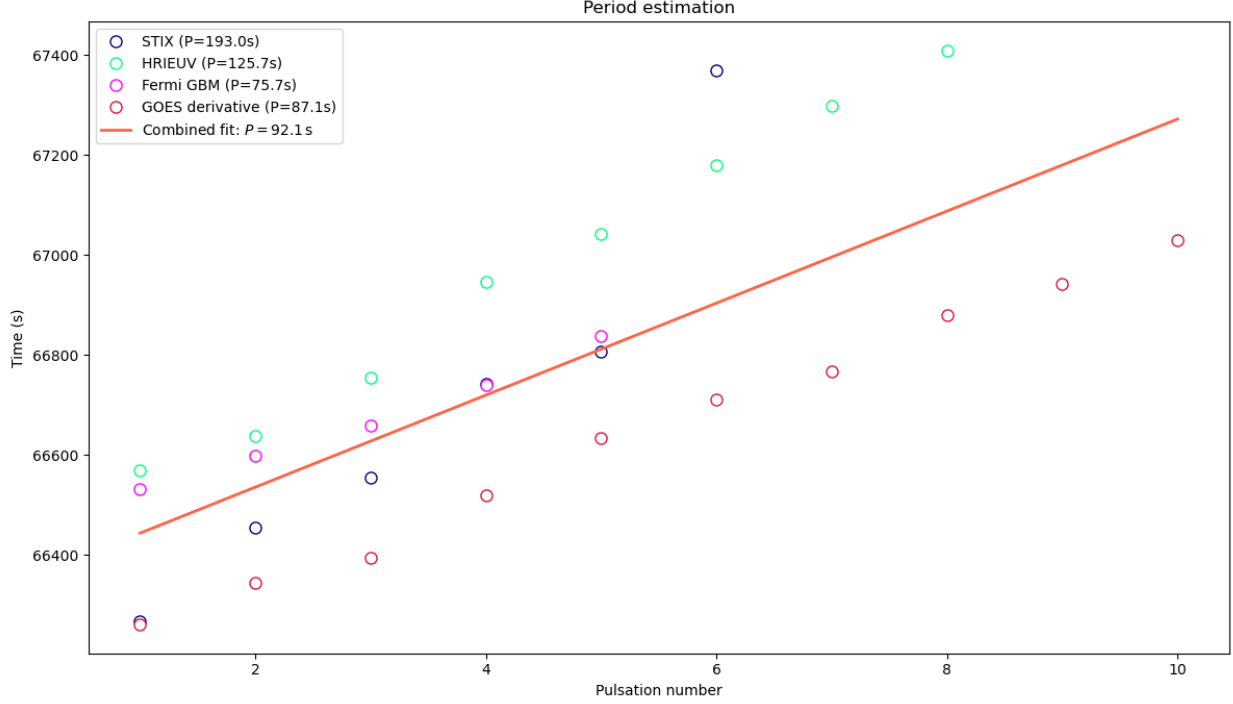
In the similar area to that where black kernel was located, we made multiple (81) smaller kernels (area: $(30'')^2$), with overlap of $15''$ on all sides. We extracted light curves from all of those and made animation: *how light curves are changing shape when different kernel is sourced*. We can't show the animation here. Extension of that part of analysis was the closer inspection of those HRIEUV light curves. We found which one correlates with multi-wavelength light curves the most, and we used that light curve to determine periodicities (green one in the resumption).



Light curves from various instruments



Light curves on the previous plot (big one) are time-corrected and they're trimmed to the same time range. Periodicities are examined via **Wavelet**, **Lomb-Scargle** and **manual** method. In manual method we visually identify peaks. Actually, we only show those results, since we're unsure about first two methods performance.



Additional remarks

1. We successfully detected quasi-periodic pulsations with a period of 92.1s across four independent instruments (STIX, EU/HRIEUV, Fermi/GBM, GOES). We're not sure about consistency here, and STIX light curve wasn't dealt with in the possible way. Period falls within expected range for M-class flares (this is broad statement). We need to confirm our findings with Lomb-Scargle and Wavelet approach.
2. This period suggests MHD oscillations in the flare arcade (slow-mode waves maybe), and periodic magnetic reconnection modulating particle acceleration. However, we're not ready to speculate about loop lengths.
3. Another idea is to compare early impulsive vs. peak phase period, because this could reveal changes in the magnetic configuration.

APPENDIX: period estimation (manual method)

